Note on Elastic Space

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Abstract: Personal Study Notes on Works Concerning Elastic Space Models or the "Physics as Geometry" Paradigm

https://github.com/ZKZOU/Note-on-Elastic-Space

It carries a fairly strong inclination toward an elastic-space ontology on my part, so readers are encouraged to read it critically:

1. The theory of the elastic aether seems to have been quite popular for a time in the 19th century. Lord Kelvin proposed the vortex atom model based on ideal fluids. Vortex structures give rise to emergent dynamical elasticity and exerted enormous influence at the time. Lord Kelvin. (Sir William Thomson)(1867). On Vortex Atoms.Proceedings of the Royal Society of Edinburgh.

2. In 1870, Clifford first proposed that matter itself should also be the curvature of spatial geometry (3D), before the Michelson–Morley experiment. The Clifford algebra he invented remains, to this day, the underlying algebraic structure of spinor representations. Clifford, W.K.(1976). On the Space-Theory of Matter. In: Čapek, M. (eds) The Concepts of Space and Time. Boston Studies in the Philosophy of Science, vol 22. Springer, Dordrecht. https://doi.org/10.1007/978-94-010-1727-5_50 (Original work published 1876).

3. The mainstream setting of elastic aether theory at the time was either a fluid model or a matter–space dualism, so what the Michelson–Morley experiment actually falsified was the matter–space dualism or the fluid model; it did not falsify elastic

aether theory.

4. Clifford's idea that matter is also spatial geometry can directly explain why the Michelson–Morley experiment yielded its result.

5. The physical picture of the electromagnetic field vortex model that Maxwell gave before presenting Maxwell' s equations is, in fact, an aether-dynamics degree-of-freedom branch of elastic aether. J. C. Maxwell. LI.(May 1861) On physical lines of force. in The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science, https://doi.org/10.1080/14786446108643067

6. Einstein at one point more or less acknowledged that the spacetime of general relativity is essentially a new kind of aether. A. Einstein.(1920).Aether und Relativitaetstheorie. Springer Berlin, Heidelberg. https://doi.org/10.1007/978-3-642-64927-1

7. The reason contemporary physics does not accept elastic aether is that discrete aether lattice models are incompatible with Lorentz covariance, which is based on spatial isotropy. But this idea actually remains open to discussion. Let me give an example: an elastic sphere is composed of elastic molecules, yet at the human observational level it exhibits nearly spatially isotropic properties—that is, its physical properties are nearly spherically symmetric. The order-of-magnitude gap between the molecular level and the observational level in this example is far smaller than the order-of-magnitude difference between Planck-scale lattice points and the currently human-observable level. The enormous order-of-magnitude difference and randomly dislocated lattice arrangements are sufficient to achieve observational-level isotropy through statistical averaging.

8. Modern elastic space models mostly adopt an elastic solid framework, treating matter as localized deformation wavepackets in this medium. The speed of light is the maximum propagation speed on this elastic medium that is space itself; hence the speed of light cannot be superimposed, giving a self consistent explanation for the

Michelson Morley result.

9. Pure geometry in GR makes it difficult to define deformation energy, which makes it hard to explain where the energy lost by gravitational redshift and electromagnetic waves goes. But under the elastic space model, this can be explained very intuitively: the energy lost to redshift is converted into elastic potential energy. This is the advantage of elastic space, namely that it can construct global energy conservation, and an elastic space composed of discrete elastic lattice points can naturally avoid problems such as singularities, divergences, and ultraviolet divergences derived from continuous models.

10. Under the elastic space model, the essence of mass—the geometric elastic duality between mass and gravity is self-evident: To explain the essence of mass, we begin with how mass generates gravity. If gravity is a deformation of spatial geometry, and mass can induce the generation of gravity, then mass is very likely also a deformation of space. If gravity is a tensile restoring potential, then what induces the tension should be a compressive deformation. Therefore, mass is very likely a compressive geometric deformation. This kind of mass–gravity geometric duality picture is something GR has not provided.

In recent years, more and more authors have recognized this modern version of the 1800s elastic aether, namely elastic space. Most of these modern versions share one thing in common: they explain the non-superposability of the speed of light in the Michelson–Morley experiment through Clifford's proposal that matter itself is also spatial geometry. More and more authors, including some from mainstream physics circles or from some of the engineering fields, recognize elastic space and the ontological understanding that matter is spatial geometry. This shows that this ontological understanding is attractive.

Some critics of the elastic space paradigm may mention that traditional elastic models require a background. In fact, this can be explained easily: if space itself is what elastic dynamics is about, then, like the GR model, no background needs to be introduced.

However, elastic space models, elastic solid models, also have many unresolved problems in explaining physics. The topological structure of space, dynamical processes, and physical properties of space are not yet understood.

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